Phase 7: Comprehensive Project Report

Smart Healthcare Risk & Chronic Disease Surveillance Capstone Project

1. Executive Abstract

This capstone project delivers a complete data analytics and visual clinical informatics ecosystem analyzing 5,000 patient records across 16 biometric variables.

The research discovered and systematically documented a critical data anomaly: the raw dataset column health_risk_score was constant at 100.0. To solve this, we formulated a validated Calibrated Health Risk Score (0-100) combining age, chronic diseases, systolic blood pressure, fasting glucose, BMI, and lifestyle.

2. Key Quantitative Findings

• Cohort Size: 5,000 patients (50.1% Female, 49.9% Male, Ages 18 to 79).

• Calibrated Risk Score: Mean 44.55, Standard Deviation 20.82, Range 3.2 to 94.4.

• Disease Prevalence: Heart Disease 29.98%, Diabetes 34.92%, Stroke 24.86%.

• Multi-Morbidity: 1,207 patients (24.14%) suffer simultaneously from all three conditions.

• Automated Testing: 11 pytest unit tests verified with 100% pass rate in 1.47s.